2 Soln: Each line of R^n does not necessarily pass through the zero vector, hence it is not a subspace since it violates condition (a) of the definition of subspace. ∴ The given statement is False.

c) The dimension of Col A is the number of pivot columns of A.

Soln: By defn, dimension of Col A is the number of vectors in the basis of Col A. By Theorem 13, the basis of Col A are the pivot columns of A. ∴ The given statement is True.

d) The dimensions of Col A and Nul A add up to the number of columns of A.

Soln: By defn, rank A = dim Col A.
By Theorem 14, rank A + dim Nul A = number of columns of A.
∴ The given statement is True.

e) If a set of p vectors spans a p-dimensional subspace H of R^n, then these vectors form a basis for H.

Soln: By Theorem 15, the given statement is True.

18) a) If B is a basis for a subspace H, then each vector in H can be written in only one way as a linear combination of the vectors in B.

Soln: Let, $\{v_1, v_2, \ldots, v_n\}$ be B.
Let, $x \in H$. ∴ $\exists c_1, c_2, \ldots, c_n \in R$. s.t. $x = c_1v_1 + c_2v_2 + \ldots + c_nv_n$
Let, $\exists d_1, d_2, \ldots, d_n \in R$ s.t. $x = d_1v_1 + d_2v_2 + \ldots + d_nv_n$
$0 = x - x = (c_1 - d_1)v_1 + (c_2 - d_2)v_2 + \ldots + (c_n - d_n)v_n$.
∴ By defn, $v_1, v_2, \ldots, v_n$ are linearly independent, $\forall i, 1 \le i \le n$, $c_i - d_i = 0$
$\Rightarrow c_i = d_i$. ∴ The given statement is True.

b) If $B = \{v_1, \ldots, v_p\}$ is a basis for a subspace H of R^n, then the correspondence $x \mapsto [x]_B$ makes H look and act the same as R^p.

Soln: On Page 157, on the last paragraph above the section "The Dimension of a Subspace", the given statement is mentioned. ∴ It is True. □